

Improvements to the monitor programs for 6809 and 6802

 flexonsbd.blogspot.com/2019/03/68096802.html

Monday, March 11, 2019

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ASSIST09
>D F805 F908

    0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F
F800 30 8D EF BE 1F 10 1F 8B 97 9D 33 84 31 8C 35 EF 0.....3.1.5.
F810 81 C6 16 34 04 1F 20 E3 A1 ED 81 6A E4 26 F6 C6 ...4.. ....j.&..
F820 0D A6 A0 A7 80 5A 26 F9 31 8D F7 D4 8E 20 FE AC .....Z&1.... ..
F830 A1 26 02 AD A4 35 84 32 8D EF 16 8D C3 4F 1F 8B .&...5.2.....0..
F840 3F 08 20 F9 01 58 02 92 02 90 02 8E 02 70 02 8A ?. ..X.....p..
F850 00 45 02 2B FF E3 02 90 02 84 02 96 02 8A 02 93 .E.+.....
F860 02 90 03 9A 02 B7 02 D2 02 BF EF 92 04 7D 01 2D .....}.-
F870 F0 BC 00 00 00 00 F0 F8 00 00 00 00 39 01 94 01 .....9...
    0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F
F880 B1 01 CB 01 C3 01 75 01 73 01 C0 01 79 00 55 01 .....u.s...y.U.
F890 7D 02 56 01 D1 6A 8D EE F7 17 02 25 EE 6A 33 5F }.V..j.....%.j3_
F8A0 0D FB 26 11 17 06 9B 50 5A 2B 0A 11 A3 A1 26 F8 ..&....PZ+....&.
F8B0 EF 6A 16 02 1E 0F FB 37 06 C1 0B 10 22 02 0F EF .j.....7...."....
F8C0 6A 58 33 8C B8 EC C5 6E CB 41 53 53 49 53 54 30 jX3....n.ASSIST0
F8D0 39 04 10 DF 97 6D 61 26 0D AD 9D EE F9 AD 9D EE 9....ma&.....
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Improvements to the monitor programs for 6809 and 6802.

We have improved the monitor programs for the 6809 and 6802.

The ROM also includes ASSIST09 and MIKBUG2, but I tend to use my custom-made monitor more often. It's not that the terminal input/output is inconvenient; I've used ASSIST09 with a full keyboard and Video-RAM, but I suppose I just feel more attached to and find it easier to use something I've made myself. So, whether it's a

6809 or a 6802, I mainly use my custom-made monitors. However, the monitor I've been using until now is just a slightly modified version of one I used over 30 years ago with a custom-made full keyboard and Video-RAM on a 6802 system, with serial input/output added. As a

result, there's a lot of wasted workspace and it's lacking in functionality. So, I've decided to take the plunge and improve my monitor.

Points targeted for improvement

(1) Unify the operation of 6809 and 6802

- Since dual boards are used by frequently switching between 6809 and 6802, it is very inconvenient if the operation is different.

(2) Improve usability

- Instead of entering a single character command and then entering parameters, make it possible to enter them all at once. (Interactive input is just annoying once you get used to it.)
- When entering parameters, no matter how many digits you enter, only the last 4 digits (for address values) or 2 digits (for data values) will be taken as valid values. (This is a similar function to ASSIST09 and is quite convenient.)

(3) Add functions

- Make it possible to set BreakPoints. I don't think 8 like ASSIST09 is necessary, but it seems that having at least one is still convenient. However, since it will be replaced with SWI code, it can only be set in the RAM area.
- Add a checksum every 16 bytes to the Dump command. Accordingly, even if the specified address is not a round number, it will be displayed within that range. (This was actually the most time-consuming part. It was necessary for the CS display.)

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COM3:38400bps - Tera Term VT
ファイル(F) 編集(E) 設定(S) コントロール(O) ウィンドウ(W) ヘルプ(H)
ASSIST09
>D F805 F908

 0 1 2 3 4 5 6 7 8 9 A B C D E F
F800 30 8D EF BE 1F 10 1F 8B 97 9D 33 84 31 8C 35 EF 0.....3.1.5.
F810 81 08 16 34 04 1F 20 E3 A1 ED 81 6A E4 26 F6 06 ...4... ..j.&..
F820 0D A6 A0 A7 80 5A 26 F9 31 8D F7 D4 8E 20 FE AC .....Z&.1.... ..
F830 A1 26 02 AD A4 35 84 32 8D EF 18 8D C3 4F 1F 8B ..&...5.2.....0..
F840 3F 08 20 F9 01 58 02 92 02 90 02 8E 02 70 02 8A ?. ..X.....p..
F850 00 45 02 2B FF E3 02 90 02 84 02 96 02 8A 02 93 .E.+.....
F860 02 90 03 9A 02 B7 02 D2 02 BF EF 92 04 7D 01 2D .....}.-
F870 F0 BC 00 00 00 00 F0 F8 00 00 00 00 39 01 94 01 .....9...
 0 1 2 3 4 5 6 7 8 9 A B C D E F
F880 B1 01 CB 01 C3 01 75 01 73 01 C0 01 79 00 55 01 .....U.s...y.U.
F890 7D 02 56 01 D1 6A 8D EE F7 17 02 25 EE 6A 33 5F ].V..j.....X.j3_
F8A0 0D FB 26 11 17 06 9B 50 5A 2B 0A 11 A3 A1 26 F8 ..&....PZ+...&.
F8B0 EF 6A 16 02 1E 0F FB 37 06 C1 0B 10 22 02 0F EF .j.....7.....
F8C0 6A 58 33 8C B8 EC C5 6E CB 41 53 53 49 53 54 30 jX3....n.ASSIST0
F8D0 39 04 10 0F 97 6D 81 26 0D AD 9D EE F9 AD 9D EE 9....mak.....
F8E0 FB 30 8C E5 3F 03 9E F8 27 0D 6F 02 6F 03 CC 01 .0..?...'..o.o...
F8F0 A6 A7 01 E7 84 6F 01 3F 06 17 06 46 2A 0C 5D D7 .....o.?...F*.P.
 0 1 2 3 4 5 6 7 8 9 A B C D E F
F900 FA 5A 2B 06 A6 30 A7 B1 20 F7 AE 6A 9F 93 86 3E .Z+..0.. ..j...>

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In the case of ASSIST09

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COM3:38400bps - Tera Term VT
ファイル(F) 編集(E) 設定(S) コントロール(O) ウィンドウ(W) ヘルプ(H)
M6809 MONITOR Ver6.1
>D F805 F908

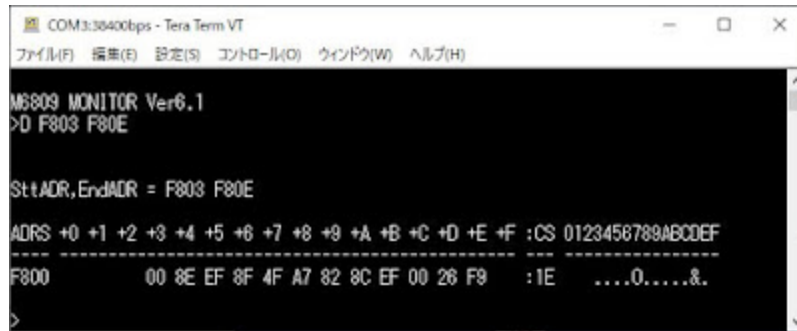
SttADR.EndADR = F805 F908

ADRS +0 +1 +2 +3 +4 +5 +6 +7 +8 +9 +A +B +C +D +E +F :CS 0123456789ABCDEF
-----
F800 EF 8F 4F A7 82 8C EF 00 26 F9 86 :18 ..0.....&..
F810 43 B7 F0 BC 12 86 15 B7 F0 BC 86 43 B7 F0 CC 12 :04 C.....C....
F820 86 15 B7 F0 CC 8E F6 38 BF EF 88 BF EF 8A BF EF :E6 .....8.....
F830 8C BF EF 00 BF EF 04 8E FE 02 BF EF 02 8E F5 FF :7C .....
F840 17 02 2D 17 02 38 86 3E 17 F7 B7 8D 19 8E F6 14 :5E ..-.8.>.....
F850 A1 84 27 09 30 03 8C F6 38 26 F5 20 E6 17 02 1E :9A ..'.0...8&....
F860 AE 01 AD 84 20 DD 8E EF 40 5F 17 F7 93 A7 80 5C :1D ....@.....Y
F870 C1 40 26 04 88 0D A7 1F 81 0D 26 EE F7 EF 3F 8E :D9 .0&.....&...?.
F880 EF 40 A6 84 39 10 8E EF 37 4F C6 19 A7 A2 5A 26 :4D .@.9...70....Z&
F890 FB 8E EF 42 10 8E EF 1E A6 80 A7 A0 81 0D 27 09 :8F ...B.....
F8A0 81 20 26 F4 88 0D A7 A2 39 7F EF 3E B6 EF 3F 81 :E1 .&.....9...>..?.
F8B0 03 22 04 1A 01 20 15 8D 48 8D 1A 34 02 A6 1F 81 :71 "... ..H..4....
F8C0 0D 26 02 86 FF B7 EF 3E 35 02 1C FE 39 8D 6B 8E :AE .&.....>5...9.k.
F8D0 EF 82 8D 13 39 34 10 8E EF 8D 8D 0B 1F 89 8E EF :48 ....94.....
F8E0 82 8D 04 1E 89 35 90 A6 8D 8D 0D 48 48 48 34 :93 .....5.....HHH4
F8F0 02 A6 84 8D 03 AA ED 39 8D 30 81 0A 25 02 8D 07 :68 .....9.0...%...

ADRS +0 +1 +2 +3 +4 +5 +6 +7 +8 +9 +A +B +C +D +E +F :CS 0123456789ABCDEF
-----
F900 39 34 26 10 8E EF 84 C8 04 :8E 94&.....

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In the case of 6809MON



It works correctly even when displayed on a single line.

Difficulties

Previously, the monitor was created for the 6802 and then ported to the 6809. This time, however, it was created first for the 6809 and then rewritten for the 6802.

While simple instruction substitutions weren't a major problem, differences in the flags for equivalent instructions caused considerable difficulty. In particular, the differences in the flags for the 6800's CPX and the 6809's CMPX were a major headache. Furthermore, while single-board 6809 and 6802 systems worked without issue, dual-board 6809/6802 systems sometimes failed to boot 6800FLEX. Once booted, it continued to function normally, so the hardware wasn't the problem. I suspected the FLEX boot command implemented in the monitor and tried various modifications, but ultimately concluded it was a hardware issue, not a software one. Currently, I'm investigating the 09/02 switching mechanism. (This took over two months.)

The revised monitor version 6.1 is now complete.

So, I decided to make the revised version of the monitor Ver6, and after using it for about half a month and making some adjustments, Ver6.1 was completed.

[illegible]

A 16KB ROM from the 27128 is sufficient, but if you use the 27256 or 27512, arrange them as shown above. This allows you to switch between your custom monitor and ASSIST09 or MIKBUG2.

. I also modified FlexDrvWin.exe, the virtual drive system used to run FLEX. I improved how Read and Write commands are received, and to allow users to customize their environment, I made it possible to read an ini file placed in the same directory at startup and set its values (port number, baud rate, handshake, window position). I'll upload my ini file FlexDrvWin.ini to [OneDrive](#) . Note that this was built with Visual Studio 2013 Community, so you may need that runtime.